

Volatility Forecasting: GARCH and Recurrent Models

Research design, historical diagnostics,
and validation review

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REVIEW STATUS

Historical predictions were independently checked and evaluated on common dates. The GARCH implementation and target transformation still require correction. No models were retrained for this review, and comparative performance is not validated.

Research question

Can recurrent models add useful information to a parsimonious conditional-variance model? This study compares a Student-t GARCH(1,1) baseline with LSTM, GRU, GARCH-feature hybrids, and additive residual hybrids on daily S&P 500 returns. It also examines what recurrent gates reveal about fitted model behavior.

What the evidence currently supports

Evaluation design matters. Seven saved prediction series cover different date ranges. The revised comparison uses the same 4,473 dates, from March 8, 2007 to December 11, 2024.

Low squared error is not sufficient. Residual hybrids have relatively low historical MSE but hit the $1e-12$ variance floor on 99 and 297 dates, respectively. Their QLIKE penalties expose a material underprediction failure.

The benchmark must be repaired. Reusing a fitted GARCH result without updating its state repeats forecasts between refits. This compromises both the baseline and models built from it.

Gate behavior is descriptive. A negative LSTM forget-gate/VIX association is an observable pattern, not proof of an adaptive memory mechanism or a tradable signal.

Scope: one index; historical daily forecasts; no trading strategy, transaction-cost study, statistical significance test, or fresh training run. Original experiment: February 2026.

01 / Data and Model Design

The data pipeline uses yfinance ticker **^GSPC** over 2003-2024. The forecast target is the squared daily log return, $y_t = r_t^2$. This is a noisy second-moment proxy; identifying it with conditional variance additionally requires a zero conditional-mean assumption.

Family	Specifications	Prediction design
GARCH	Student-t GARCH(1,1)	Zero-mean return model; parameters refit every 21 trading days. Daily state update is currently missing.
Pure recurrent	LSTM and GRU	21 lagged rows of log return, squared return, absolute return, and 21-day rolling realized volatility.
Feature hybrid	LSTM and GRU	Same lagged features plus the historical GARCH variance series.
Residual hybrid	LSTM and GRU	Predict the signed difference between squared return and GARCH variance, then add the predicted correction to GARCH.

Timing and validation

Each neural forecast for date t uses feature rows $t-21$ through $t-1$. The rolling split configuration has a minimum training span of 756 observations, a 252-observation validation block, and 21-observation test blocks advanced by 21. Training windows expand and overlap; test blocks do not overlap. Feature warmup and sequence construction reduce effective sample sizes.

Feature and target statistics are fitted on each training split and applied to validation and test data. The rolling-volatility input is annualized, while the forecast target and reported losses use daily variance. The feature hybrid uses lagged GARCH values; it does not append the current-date GARCH forecast directly to the recurrent output.

Neural configuration recorded in the notebooks

Lookback 21; 8 recurrent units; dropout 0.1; linear output; Adam learning rate 0.001; batch size 64; up to 35 epochs; early-stopping patience 6; seed 42 plus split index. These are notebook settings, not a completed hyperparameter search. A fixed seed alone does not ensure reproducibility across library versions or hardware.

Implementation: `src/data/pipeline.py`; `src/models/garch.py`; `src/models/rnn.py`; notebooks 02-05a. Model families total seven specifications, not six.

02 / Comparable Historical Results

All rows below use **4,473 common dates: March 8, 2007 to December 11, 2024**. Dates are aligned, but the underlying forecasts remain from the original, uncorrected runs. This is an arithmetic reconciliation, not a new out-of-sample experiment.

Saved forecast	MSE / 10 ⁻⁷	QLIKE	Floor hits
GARCH-t	3.058102	-8.085829	0
Pure LSTM	3.566812	-5.272290	0
Pure GRU	3.561952	-3.776601	0
Feature LSTM	3.602508	-4.088043	0
Feature GRU	3.572742	-2.506646	0
Residual LSTM	2.860449	858,837.12	99
Residual GRU	3.082436	4,000,312.63	297

$$\text{MSE} = \text{mean}[(y - h)^2]$$

$$\text{QLIKE} = \text{mean}[\log(h) + y / h]$$

Here y is the observed squared-return proxy and h is forecast daily variance. The implementation clips both inputs at $1e-12$ when calculating QLIKE. MSE uses the stored values without that clipping. Lower is better for both losses. This QLIKE convention can be negative; its level is not a percentage or an economic return.

How to read the disagreement

Residual LSTM has the lowest MSE in these saved outputs, yet its near-zero forecasts create a very large QLIKE penalty. This is evidence of a calibration failure in this implementation, not proof that a residual architecture is inherently inferior.

GARCH has the lowest historical QLIKE, but its stale-state implementation prevents a defensible claim of benchmark superiority. Likewise, differences among neural models are point estimates without confidence intervals or multi-seed replication.

Both MSE and QLIKE can preserve expected model rankings under suitable conditions on an unbiased noisy volatility proxy [4]. They weight errors differently and can disagree in finite samples. Neither loss alone validates a model or repairs a flawed benchmark.

Source: seven locally retained prediction CSVs; independent NumPy recomputation. Saved native-sample metrics reconcile to rounding. Native coverage is documented in the appendix.

03 / Positivity and Loss Geometry

The residual model predicts $u_t = y_t - g_t$, where g is the saved GARCH forecast. This is a signed variance-proxy difference, not a standardized return innovation. The reconstructed forecast is $h_t = \max(g_t + \text{predicted } u_t, 1e-12)$.

Residual LSTM reaches the floor on **99 / 4,473 dates (2.21%)**; Residual GRU on **297 / 4,473 (6.64%)**. When y is positive and h approaches zero, the y/h term dominates QLIKE. A small MSE does not rule out this failure.

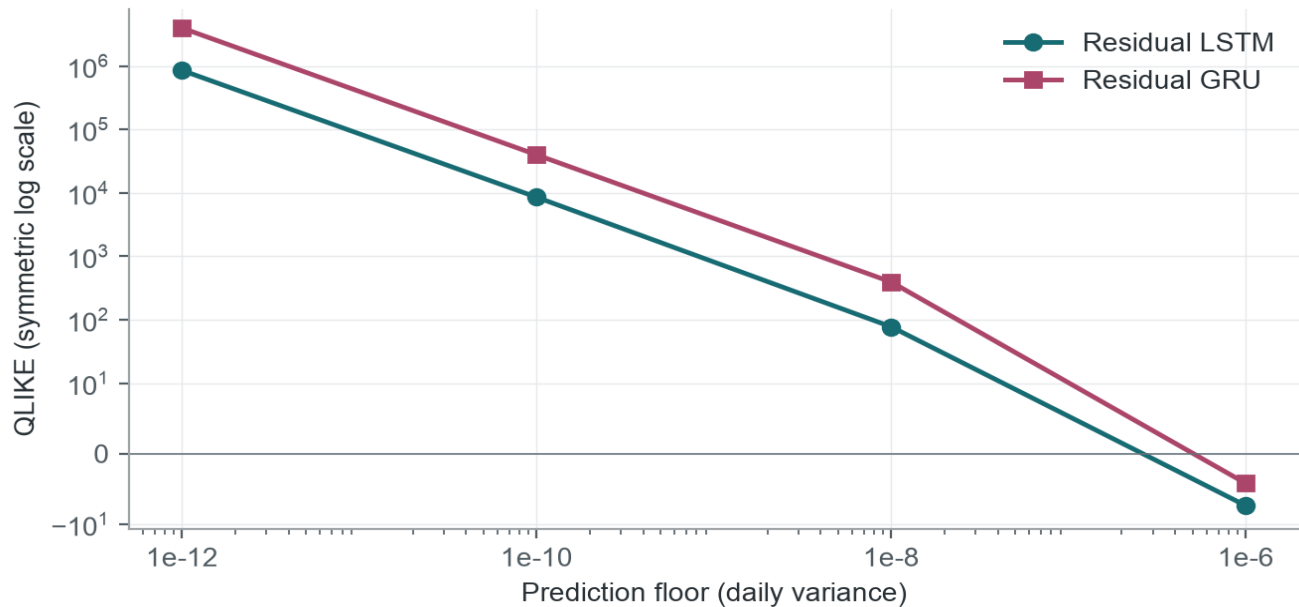


Figure 1. Sensitivity of the same saved residual forecasts to the prediction floor. Only the prediction floor changes; the realized-proxy floor stays at 1e-12. The vertical axis uses a symmetric log scale. This is not a retuned model or a validated floor selection.

A separate target-transform mismatch

The non-residual target transform uses $\log(\max(y, 1e-8))$, but its inverse subtracts 1e-8 after exponentiation. That subtraction does not invert the clipping transform. A round-trip check maps $y = 1e-6$ to approximately 9.90e-7, and $y = 1e-8$ to approximately zero.

Even with a consistent inverse, minimizing squared error on log targets and exponentiating does not generally recover the conditional mean variance. A revised experiment should separate transformation correctness, objective alignment, and positivity constraints. A multiplicative correction or a positive output link is a candidate to test, not a proven fix.

04 / What the Gates Can Tell Us

The diagnostic averages saved recurrent activations across units and lags 1-20 for each forecast date, then relates the aggregate to the same-date VIX close. This is an **ex-post descriptive comparison**: that VIX close is not available at the previous-day forecast origin and is not presented as a predictive feature.

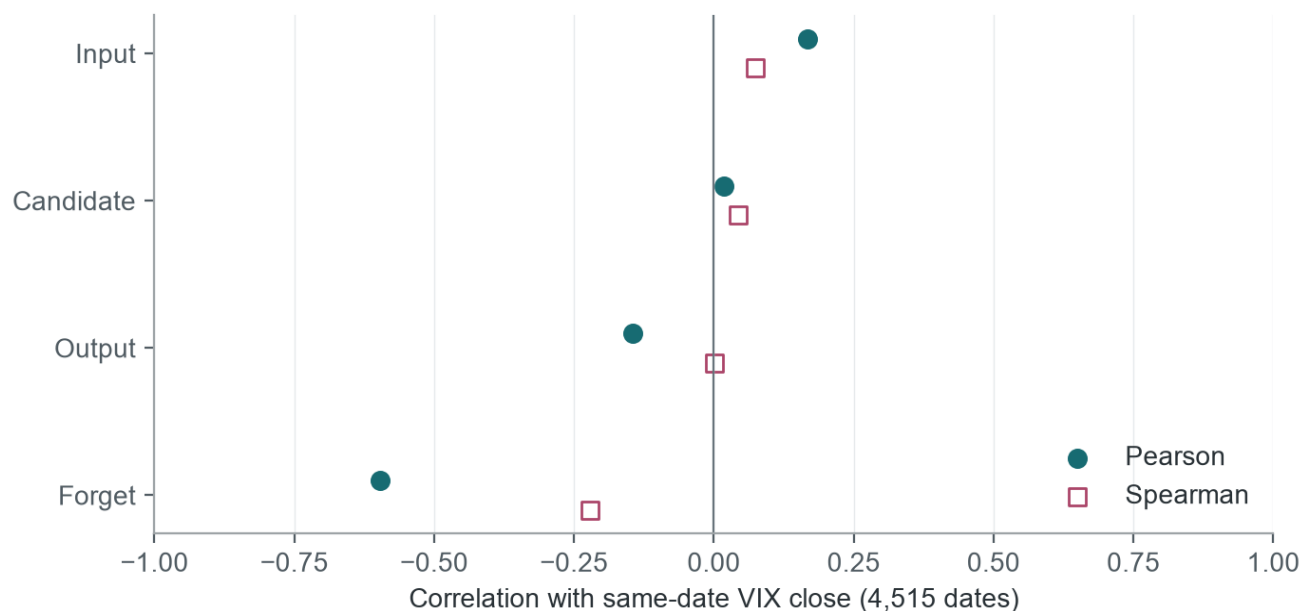


Figure 2. Pure-LSTM gate summaries versus VIX, January 5, 2007 to December 11, 2024 (4,515 dates). Pearson and Spearman are recomputed from the retained per-date summaries. No confidence intervals have been estimated.

The strongest linear association in this example

The pure-LSTM forget-gate aggregate has Pearson correlation **-0.596** and Spearman correlation **-0.221** with VIX. The difference between these measures is a reason to inspect outliers, nonlinearities, and regime dependence, rather than claim a stable monotone relationship.

Why this is not a mechanism test

The time series stitches together separately trained models across rolling splits. Unit identities need not be stable, and an average forget-gate activation is not an identified memory horizon. Cross-architecture correlations also do not isolate the effect of a gate from training, features, or forecast calibration.

Next tests should include fold-level and regime-level results, lagged VIX comparisons, multiple seeds, and controlled ablations. Any predictive test must respect the forecast information set. No signal alpha or causal interpretation is asserted here.

05 / Conclusions and Next Experiment

The strongest contribution of the current artifact is a transparent research workflow: temporal model evaluation, scale-aware losses, constrained versus unconstrained forecast construction, and diagnostics that reveal implementation failures. It does not yet establish a superior forecasting model.

Priority	Correction or test	Acceptance evidence
1 / Blocking	Update GARCH state with every new return between parameter refits.	A regression test shows a new return changes the next forecast with parameters fixed; regenerate all dependent features.
2 / Blocking	Make target transform and inverse consistent; enforce positive forecasts by construction.	Round-trip tests above the clipping floor; explicit behavior at and below the floor; no unexplained clipping failures.
3 / Required	Rerun all seven specifications on identical test dates and information sets.	Same-date prediction bundle, coverage checks, timestamp audit, and reproducible environment/configuration.
4 / Required	Quantify uncertainty and calibration.	Multiple seeds, paired loss differences with time-series-appropriate inference, tail/regime diagnostics, and simple persistence/EWMA baselines.
5 / Extension	Test whether gates add explanatory or predictive value.	Pre-specified lagged diagnostics and controlled ablations; not just pooled contemporaneous correlation.

Training diagnostics need a narrower interpretation

The existing training log stores the minimum training loss and minimum validation loss separately. Those minima may come from different epochs, so their difference is not a paired checkpoint generalization gap. Log-variance and signed-residual training objectives are also on different transformed scales. Neither comparison supports a claim that one architecture is free of overfitting.

Readiness

Suitable: a portfolio discussion of research design, implementation review, and failure analysis.

Not yet suitable: a validated model ranking, production risk model, or evidence of a profitable trading strategy.

06 / References and Provenance

- [1] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307-327.
[https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1)
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- [3] Cho, K. et al. (2014). Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation.
<https://arxiv.org/abs/1406.1078>
- [4] Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246-256.
https://public.econ.duke.edu/~ap172/Patton_vol_proxies_JoE_2011.pdf
- [5] arch documentation. Univariate volatility forecasting: forecast origins, horizons, and model assumptions.
<https://arch.readthedocs.io/en/latest/univariate/forecasting.html>

Evidence boundary

The review checked all seven locally retained forecast series for duplicate dates, non-finite values, target agreement, sample coverage, and metric reconciliation. It also recomputed 21 architecture/variant/gate correlations, inspected the relevant source code and notebook settings, and ran focused control-flow and transform checks. It did not download a fresh market-data snapshot or retrain the models.

The public repository contains code and aggregate review data, not the underlying prediction CSVs or licensed market-data snapshots. The aggregate evidence file includes SHA-256 hashes of the reviewed local prediction files and Git blob hashes of the reviewed code. These hashes identify artifacts; they do not substitute for an independently reproducible full training run.

Repository and companion material

github.com/SRG-eddieliu/rnn-volatility-lab

Companion: Volatility Forecasting - Diagnostic Appendix.

Aggregate evidence: `reports/review_metrics.json`.

Validation status: `docs/validation-status.md`.

Research provenance

The underlying experiments originated in a collaborative research project. This revised portfolio report is presented by Eddie Liu. Contributor names and course identifiers have been omitted; the byline does not imply sole authorship of the original experiments.

Revision note: this document replaces the earlier narrative with corrected method descriptions, common-date arithmetic, explicit implementation limitations, and references. It does not change the underlying model code or claim a corrected experimental result.

Volatility Forecasting: Diagnostic Appendix

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A / Coverage and Reconciliation

This appendix documents checks of retained historical outputs. It is not a fresh model run. All comparative losses use the 4,473-date intersection; gate tables retain each variant's native diagnostic coverage.

Forecast series	Native n	Start	End
GARCH-t	4515	2007-01-05	2024-12-11
Pure LSTM	4515	2007-01-05	2024-12-11
Pure GRU	4515	2007-01-05	2024-12-11
Feature LSTM	4473	2007-03-08	2024-12-11
Feature GRU	4473	2007-03-08	2024-12-11
Residual LSTM	4473	2007-03-08	2024-12-11
Residual GRU	4473	2007-03-08	2024-12-11

What was checked

All seven series have unique prediction dates, finite targets and forecasts, nonnegative targets, and strictly positive stored forecasts. Native-sample MSE and QLIKE reproduce the saved metric file to numerical rounding. Positive stored forecasts do not rule out clipping failures: residual forecasts include values exactly at $1e-12$.

On common dates, the largest absolute difference among stored targets is $5.2e-10$. Targets pass a relative tolerance of $1e-6$ with absolute tolerance $1e-12$, consistent with floating-point rounding. The review retains each file's stored target rather than silently replacing it.

Input scope: yfinance ^GSPC, 2003-2024 including warmup and training. The historical evaluation period does not begin in 2003. Prediction timestamps, corporate data revisions, and full experiment reproducibility still require a fresh end-to-end audit.

B / Historical Loss Comparison



Figure A1. Historical common-date MSE for all seven specifications. Bars start at zero; values are scaled by 10^{-7} . The model order is by family, not a validated ranking.

Saved forecast	MSE / 10 ⁻⁷	QLIKE	Floor hits
GARCH-t	3.058102	-8.085829	0
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Feature GRU	3.572742	-2.506646	0
Residual LSTM	2.860449	858,837.12	99
Residual GRU	3.082436	4,000,312.63	297

All rows: $n = 4,473$, March 8, 2007 to December 11, 2024. Lower is better for both losses. QLIKE is shown in a table rather than compressing negative and million-scale positive values onto a misleading linear comparison plot.

Interpretation limit: matching dates fixes unequal coverage in the table, but does not correct stale GARCH forecasts, transform inconsistencies, or missing uncertainty estimates.

C / GARCH State-Update Audit

The baseline repeatedly calls forecast(horizon=1) on the same fitted result between parameter refits. No new return is supplied to that result. The returned one-step forecast remains anchored at the fitted endpoint rather than advancing one day at a time.

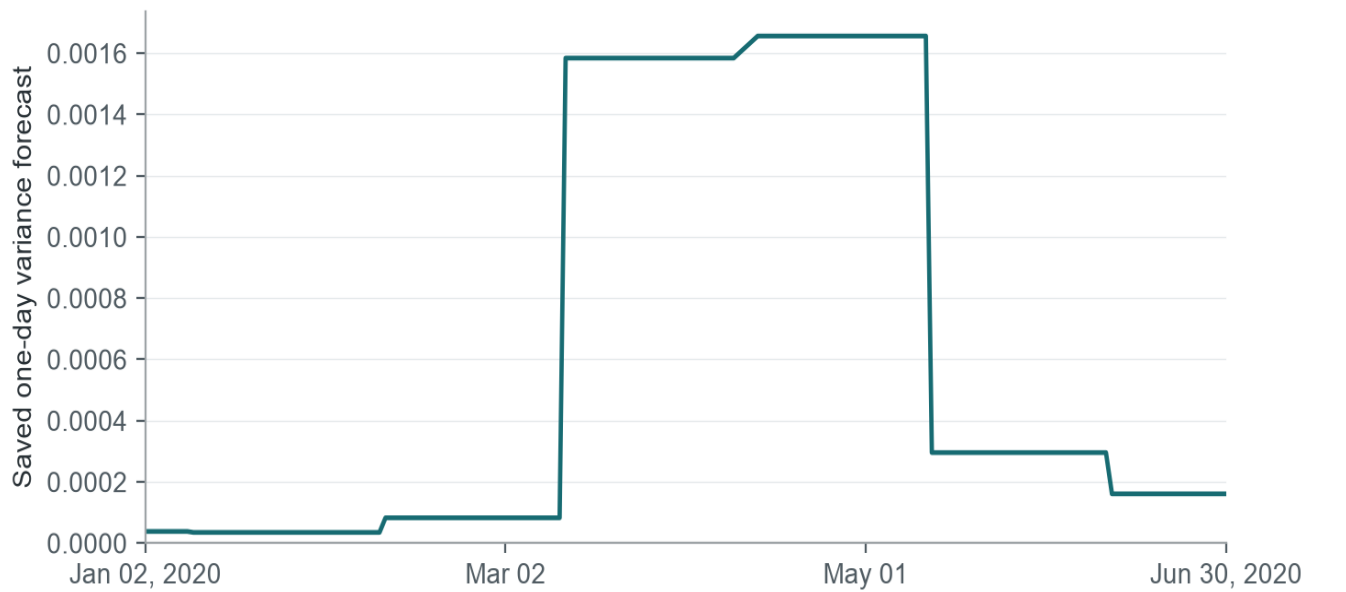


Figure A2. A six-month slice of saved baseline predictions. This plot is selected to show repeated forecasts, not to estimate comparative performance.

Whole-series check	Observed result
Identical to the previous forecast	4,300 of 4,514 transitions (95.26%)
Median / maximum constant run	21 / 21 trading observations
Focused control-flow check	With fitting mocked, changing a return between refits leaves the tested next forecast unchanged.

Why parameter refitting and state updating differ

GARCH parameters may be held fixed for several days while the conditional variance state is updated after each newly observed return. Re-estimation every 21 days is not itself the defect. Reusing the endpoint forecast without the intervening state recursion is the defect. The mock test isolates this control flow; it does not validate a fitted GARCH model or provide a corrected forecast.

Affected artifacts: GARCH baseline; GARCH-feature inputs; residual targets and reconstruction. These dependencies must be regenerated after correction.

D / Numerical and Objective Checks

Floor sensitivity

Prediction floor	Residual LSTM QLIKE	Residual GRU QLIKE
1e-12	858,837.115884	4,000,312.632582
1e-10	8,580.029944	39,997.212392
1e-08	77.559990	394.360908
1e-06	-7.384414	-4.224426

Prediction-only floor sensitivity on 4,473 common dates. The observed-target floor remains 1e-12 for every row. A larger floor mechanically reduces severe underprediction penalties; choosing a floor after inspecting test losses would introduce selection bias.

Target transformation

Implemented forward target map:

$$z = [\log(\max(y, \text{eps})) - \text{train_mean}] / \text{train_std}$$

Implemented inverse:

$$h = \exp(z * \text{train_std} + \text{train_mean}) - \text{eps}$$

$$\text{eps} = 1\text{e-}8$$

Input y	Round-trip output (approximately)
1e-8	0 (small floating-point error before clipping)
1e-6	9.90e-7
1e-4	9.999e-5

A clipping transform is not invertible below its floor. Above the floor, however, subtracting eps introduces an avoidable discrepancy. Any correction must explicitly define behavior at the floor and test the transformed training objective against the forecast estimand.

Checkpoint diagnostics

The existing train/validation gap uses separately minimized losses, potentially from different epochs. A revised log should evaluate both samples at the selected checkpoint. Losses across log-target and signed-residual objectives should not be compared as if they shared a common scale.

E / Gate Diagnostics: Full Summary

Same-date VIX close versus unit-averaged, lag-1-to-20 activations. All 21 Pearson correlations reconcile to the saved summary; Spearman values are independently computed from ranks. Native coverage differs across variants.

Model	Activation	n	Pearson	Spearman
Pure LSTM	candidate	4515	+0.018	+0.043
Pure LSTM	forget	4515	-0.596	-0.221
Pure LSTM	input	4515	+0.167	+0.075
Pure LSTM	output	4515	-0.145	+0.000
Feature LSTM	candidate	4473	-0.098	+0.041
Feature LSTM	forget	4473	-0.322	-0.170
Feature LSTM	input	4473	+0.082	-0.006
Feature LSTM	output	4473	-0.265	-0.098
Residual LSTM	candidate	4473	-0.127	-0.012
Residual LSTM	forget	4473	-0.104	+0.231
Residual LSTM	input	4473	+0.362	+0.277
Residual LSTM	output	4473	+0.139	+0.324
Pure GRU	candidate	4515	+0.023	+0.074
Pure GRU	reset	4515	+0.227	+0.179
Pure GRU	update	4515	+0.114	+0.277
Feature GRU	candidate	4473	+0.059	+0.105
Feature GRU	reset	4473	+0.062	-0.062
Feature GRU	update	4473	+0.088	+0.267
Residual GRU	candidate	4473	+0.024	+0.104
Residual GRU	reset	4473	+0.270	+0.232
Residual GRU	update	4473	-0.026	+0.078

Candidate activations are included with the gate summaries for consistency with the saved files; they are not sigmoid gates. These are pooled correlations across rolling refits, without uncertainty estimates, adjustment for serial dependence, or a causal interpretation.

F / Verification and Provenance

Category	Observed defects / evidence	Assessment
Forecast calculation	Stale GARCH state; target inverse mismatch.	Not ready for validated model comparison.
Evaluation	Seven native metric rows reconcile; all seven aligned to 4,473 dates. Residual floor failures persist.	Arithmetic checked; model validity unresolved.
Interpretation	Original gate-mechanism and model-superiority claims exceed the available evidence.	Reframed as hypotheses and historical diagnostics.

Category	Observed defects / evidence	Assessment
Report design	Original narrative mixed unequal samples and omitted key implementation limitations.	Replaced with explicit dates, definitions, limitations, and source-backed figures.
Completeness	No fresh training, repeated seeds, formal loss-difference tests, or economic backtest.	Not presented as completed validation.
Reproducibility	Public code and aggregate evidence; local prediction CSVs are not bundled.	Artifact hashes recorded; full independent rerun pending.

Source trail

Aggregate values, sensitivity outputs, focused test results, eight source-file SHA-256 hashes, and four reviewed code-blob hashes are published in `reports/review_metrics.json`. The review inspected the report narrative and relevant code; the 54-page legacy diagnostic deck was used as context, not certified page by page.

Open limitations

No fresh yfinance download; no retraining; no line-by-line audit of every repository module; no confirmation that all historical charts used the exact same artifact version. The new tables and figures use only the explicitly checked sources. The legacy report-generation script still describes the historical workflow.

Research provenance

The underlying experiments originated in a collaborative research project. This revised portfolio report is presented by Eddie Liu. Contributor names and course identifiers have been omitted; the byline does not imply sole authorship of the original experiments.